Chapter 11, Problem 91CP

Problem

The nameplate of an electric motor has the following information:

Line voltage: 220 V rms

Line current: 15 A rms

Line frequency: 60 Hz

Power: 2700 W

Determine the power factor (lagging) of the motor. Find the value of the capacitance *C* that must be connected across the motor to raise the pf to unity.

Step-by-step solution

Step 1 of 1

$$P = S\,Cos\,\theta$$

$$Pf = Cos\,\theta = \frac{P}{S} = \frac{2700}{(220)(15)} = 0.8182$$

$$Q = S\,Sin\,\theta = 220(15)\,Sin(35.09°) = 1897.3$$

When the power is raised to unity Pf , $Q_{120°}$.

And, $Q_C = Q = 1897.3$

$$C = \frac{Q_L}{\omega V_{rms}^2} = \frac{1897.3}{(2\pi)(60)(220)^2}$$

$$= 104\,\mu F$$.